

CSE218 Final Quiz

19 February 2022

Full Marks: 40, Time: 40 mins (answer: 35 mins and scan and upload: 5 mins)

Instructions:

- i. There are eight (8) questions. Each question carries 5 marks.
- ii. Answer all the questions.
- iii. Answer each question in a separate page. Do not mix answers of multiple questions in the same page.

Answer all the questions below.

1. Use Gauss elimination with partial pivoting to solve the following system of linear equations.

$$4x_1 + x_2 - x_3 = -2$$

$$5x_1 + x_2 + 2x_3 = 4$$

$$6x_1 + x_2 + x_3 = 6$$

2. Find the root of the equation $x^2 = 0.5 + \sin(x)$ in the interval $[-1, 0]$. Use the interval endpoints as the initial guesses. Show the first 5 iterations of the bisection method. For each iteration, show the estimate of the root and absolute approximate relative error.
3. Find the approximate value of $\sqrt[3]{10}$ using Newton-Raphson method. Use 3 as the initial guess. Show the Newton-Raphson update equation for the given problem. Show the estimate of the root and absolute approximate relative error for the first three iterations.
4. Consider the given dataset below. Now -

- Find the value of $f(x)$ using the **Newton's divided difference method for quadratic interpolation** where,

$$x = 12 + 3 * ((\text{last three digit of your student no} - 1) \% 30)$$

As an example, if your student no is 1905125, the value of x will be,

$$x = 12 + 3 * ((125 - 1) \% 30) = 24$$

- Find the **absolute approximate relative error** for the $f(x)$ value computed above.
- Show detailed calculation steps including the divided difference table.

x	f(x)		x	f(x)
10	28.33		58	6.25
13	38.22		61	20.13
16	22.8		64	50.56

CSE218 Final Quiz
19 February 2022

Full Marks: 40, Time: 40 mins (answer: 35 mins and scan and upload: 5 mins)

19	30.36		67	24.66
22	37.98		70	6.36
25	57.25		73	74.19
28	41.87		76	82.23
31	55.67		79	13.8
34	33.92		82	70.48
37	92.81		85	66.58
40	80.46		88	16.65
43	29.46		91	77.41
46	32.08		94	41.41
49	72.48		97	88.66
52	62.51		100	62.43
55	79.68		103	40.51

5. Compare **Newton's divided difference** and **Lagrangian** methods of interpolation in terms of their respective advantages and disadvantages.

6. You need to find the n -th term of the following sequence:

14, 68, 158, ...

It is given that the term will be a polynomial expression. Which numerical method will you use to solve the problem? Show your work step by step.

7. Use Simpson's rule with $n = 8$ to approximate the integral below

$$I = \int_2^6 f(x)$$

A table of values for the function $f(x)$ is given below.

x	1.5	1.75	2	2.25	2.5	2.75	3	3.25	3.5	3.75
$f(x)$	1.58	3.06	5.04	7.6	10.75	14.54	18.97	24.01	29.63	35.73

x	4	4.25	4.5	4.75	5	5.25	5.5	5.75	6	6.25
$f(x)$	42.16	48.72	55.12	60.94	65.63	68.48	68.5	64.43	54.63	36.94

CSE218 Final Quiz
19 February 2022

Full Marks: 40, Time: 40 mins (answer: 35 mins and scan and upload: 5 mins)

8. You have to examine the relationship between the age and price for used cars sold in the last year by a car dealership company. Here is the table of the data:

Car age (in years)	Price (in dollars)
4	5800
5	5700
7	4200
8	4100
9	3100
10	2500

Which model fits the data best? Find the corresponding coefficients and hence the equation.